

SAT In Terms Of Math Problems

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1-4

“In Terms of” questions ask you to express a certain variable “in terms of” the other variables. If a question says, “solve for x in terms of y and z ,” you need to isolate x and determine what it is equal to. If you have done this correctly, the x will be isolated on one side the equation and the y and z values, along with any numbers, will be on the other side of the equation.

The expressions in these questions can range from very simple to extremely complex and intimidating, but as long as you use the algebra skills that we learned in Chapter 4, you should be able to solve these questions without any issue. The answer choices usually make it obvious which variable you are being asked to isolate.

Example 1: A painter uses the formula $z = 20hw$ to estimate that z liters of paint are needed to paint a wall that is h feet high and w feet wide. Which of the following expresses w in terms of h and z ?

- A) $w = 20z - h$ B) $w = \frac{z}{h-20}$ C) $w = \frac{z}{20h}$ D) $w = \frac{h}{20z}$

Solution: The question asks for w in terms of h and z , so to solve we isolate w . To get w by itself, we divide both sides by $20h$.

$$z = 20wh$$

$$\frac{z}{20h} = w$$

The equation above can also be written as $w = \frac{z}{20h}$, so **the answer is C**.

Example 1 is an easy example of an “in terms of” questions. Now, let’s look at another example that is a bit more challenging but still not too difficult.

Example 2: The equation below describes how a water pump displaces w gallons of water from a well every h minutes when the water is at a depth of x feet. Which of the following equations expresses x in terms of h and w ?

$$2000 + xh^2 = 4wh$$

- A) $x = \sqrt{4wh - 2000}$
B) $x = 4wh - 2000 + h^2$
C) $x = \frac{4wh - 2000}{h^2}$
D) $x = \frac{4w}{h} - 2000$

Solution: We need to isolate x . To start, we want to get all terms with x on the left side and all other terms on the right side. To do so, we subtract 2000 from both sides.

$$2000 + xh^2 = 4wh$$

$$xh^2 = 4wh - 2000$$

Now, we can divide both sides by h^2 to solve for x .

$$x = \frac{4wh - 2000}{h^2}$$

The answer is C.

Next, let's see an example that is more challenging. The SAT can make "in terms of" questions more difficult by asking you to do more advanced algebra to isolate a variable.

Example 3: For the equation below, which of the following expresses m in terms of p and q ?

$$15p - 2m = 2mq + 300$$

A) $m = 15p + 2q + 300$

B) $m = \frac{15p+300+2q}{2}$

C) $m = \frac{300-15p}{2q}$

D) $m = \frac{15p-300}{2q+2}$

Solution: We need to isolate m . To do this, we need to move all terms with m to the left side and move all other terms to the right side. We do this by subtracting $15p$ and $2mq$ from both sides to get

$$-2mq - 2m = -15p + 300$$

Now, we need to factor out m from the terms on the left side.

$$m(-2q - 2) = -15p + 300$$

To isolate m , we divide both sides by $-2q - 2$

$$m = \frac{-15p + 300}{-2q - 2}$$

This does not match any of the answer choices yet, but we are close. If we multiply the right side by $\frac{-1}{-1}$, the equation instead will look like

$$m = \frac{15p - 300}{2q + 2}$$

The answer is D.

If the variable that you are trying to isolate appears in multiple terms, as with m in Example 3 above, we always need to move all terms including that variable to one side and then factor to isolate the variable. Once the term is factored out, you can then use division to isolate the term and find the correct answer. If you need to review how to factor to isolate a variable, go to page 24.

3

1. The velocity v of an object is found by taking the change in position m of an object and dividing it by the time s . Which of the following gives time s in terms of m and v ?

A) $s = mv$

B) $s = \frac{v}{m}$

C) $s = \frac{m}{v}$

D) $s = m - v$

1

2. Due to a new tax, steel prices have recently increased. The original pre-tax price of a steel bar is p dollars. The new price of steel bar after the increase is represented by n dollars. The equation below shows this relationship. What is p in terms of n ?

$$1.18p + 40 = n$$

A) $p = 40n - 1.18$

B) $p = 1.18n - 40$

C) $p = \frac{n-40}{1.18}$

D) $p = \frac{1.18}{n-40}$

3. The given equation below relates the positive numbers b , x , and z . Which equation correctly expresses x in terms of b and z ? (2)

$$\frac{1}{9b} = \frac{13x}{z}$$

- A) $x = \frac{9bz}{13}$
 B) $x = z - 117b$
 C) $x = \frac{z}{117b}$
 D) $x = 113bz$

4. The given equation relates the positive numbers a , b , and c . Which equation correctly expresses c in terms of a and b ? (2)

$$\frac{4}{9a} = \frac{16b}{3c}$$

- A) $c = 144ab$
 B) $c = 12ab$
 C) $c = \frac{1}{144ab}$
 D) $c = \frac{1}{12ab}$

5. For the expression below, which of the following expresses p in terms of g , m , and q ? (1)

$$g = \frac{m + 3p + 5}{q}$$

- A) $p = \frac{gq+5}{m}$
 B) $p = \frac{gq-m-5}{3}$
 C) $p = \frac{gq-5}{3+m}$
 D) $p = \frac{gq+m+5}{3}$

6. The equation below relates the positive real a , b and c . Which equation correctly expresses c in terms of a and b ? (2)

$$a = b(17 - c)$$

- A) $c = \frac{a}{b} - 17$
 B) $c = \frac{b}{a} - 17$
 C) $c = -\frac{a}{b} - 17$
 D) $c = -\frac{a}{b} + 17$

7. If $5x = \sqrt{a}$, what is a in terms of x ? (1)

- A) $\sqrt{5x}$
 B) $5x^2$
 C) $25x$
 D) $25x^2$

8. The given equation below relates the numbers l , m , and k . Which equation correctly expresses l in terms of m and k ? (2)

$$18k + \frac{3l}{2} = 3m$$

- A) $l = -18k + 3m$
 B) $l = -24k + 4m$
 C) $l = -12k - 3m$
 D) $l = -12k + 2m$

9. Approximately 25% of the height of Mauna Kea is visible above the surface of the ocean. If v represents the height in meters of Mauna Kea visible above the surface of the ocean and h represents the total height in meters of Mauna Kea, including the portion below the ocean's surface, which of the following equations best approximates h in terms of v ? (2)

- A) $h = 0.25v$
 B) $h = \frac{v}{0.25}$
 C) $h = 1.25v$
 D) $h = \frac{v}{1.25}$

10. If x is 30% of y , which expression represents y in terms of x ? (2)

- A) $0.03x$
 B) $0.70x$
 C) $\frac{10}{3}x$
 D) $\frac{3}{10}x$

11. If $3\sqrt{5x} = b$, what is $5x$ in terms of b ? (2)

- A) $\frac{b}{3}$
 B) $\frac{b^2}{3}$
 C) $\frac{b^2}{9}$
 D) $15b^2$

3

12. The line $y = kx - 3$, where k is a constant, is graphed in the xy -plane. If the point (a, b) is on the line, where a and b do not equal to zero, what is the slope of the line in terms of a and b ?

- A) $\frac{a+3}{b}$
 B) $\frac{a-3}{b}$
 C) $\frac{b+3}{a}$
 D) $\frac{b-3}{a}$

3

13. The map of a country is drawn to scale where 3 inches represents 5 miles. The actual distance between two locations is x miles. Which expression represents the distance on the map, in inches, between the two locations?

- A) $\frac{5x}{3}$
 B) $\frac{3x}{5}$
 C) $3x + 5$
 D) $3x - 5$

2

14. Which of the following gives e in terms of a, b, c , and d ?

$$abd - bcd = ed - cd$$

- A) $abd - bcd - cd$
 B) $abd - bcd + cd$
 C) $ab - bc + c$
 D) $ab - bc - c$

3

15. The equation below relates the positive real numbers a, b, c , and d . Which equation correctly expresses b in terms of a, c , and d ?

$$a = b + bcd$$

- A) $b = \frac{a}{1+cd}$
 B) $b = \frac{1+cd}{a}$
 C) $b = a - 2cd$
 D) $b = \frac{a}{2cd}$

3

16. A helium machine blows up balloons such that after starting with an empty balloon, the machine will fill the balloon at a rate of c cubic inches of air every t seconds, where $c = 6t\sqrt{t}$. Which of the following gives the average rate at which the balloon is filled in terms of t ?

- A) $\sqrt{t}$
 B) $6\sqrt{t}$
 C) $\frac{6}{\sqrt{t}}$
 D) $6t$

3

17. If $2\sqrt[3]{5x} = b$, what is x in terms of b ?

- A) $\frac{b^3}{2}$
 B) $\frac{b^3}{10}$
 C) $\frac{b^3}{20}$
 D) $\frac{b^3}{40}$

3

18. If $a^2 = 16b^4$, what is b in terms of a ?

- A) $\frac{a}{4}$
 B) $\frac{\sqrt{a}}{4}$
 C) $\frac{\sqrt{a}}{2}$
 D) $\frac{\sqrt{a}}{16}$

4

19. The formula below is used to show E , the expected time to complete a marathon, where f is the fastest time to finish the race, A is the average time to finish, and s is the slowest time to finish. Which of the following correctly gives s in terms of E, f , and A ?

$$E = \sqrt{\frac{s^2 + 2f}{A}}$$

- A) $s = \sqrt{E^2 A - 2f}$
 B) $s = E\sqrt{A - 2f}$
 C) $s = \sqrt{\frac{E^2 A}{2f}}$
 D) $s = EA - 2$

20. The equation below relates the positive real numbers a, b, c , and d . Which equation correctly expresses d in terms of a, b , and c ? (4)

$$a = \frac{bc}{\sqrt{1+d}}$$

- A) $d = \left(\frac{bc}{a}\right)^2 + 1$
 B) $d = \frac{(bc)^2}{a^2 - 1}$
 C) $d = \frac{bc}{a^2} - 1$
 D) $d = \left(\frac{bc}{a}\right)^2 - 1$

21. The given equation relates the distinct positive real numbers j, k , and l . Which equation correctly expresses l in terms of j and k ? (4)

$$\frac{8j}{48k} = \frac{1}{6}\sqrt{l-17}$$

- A) $l = \sqrt{\frac{j}{k}} + 17$
 B) $l = \sqrt{\frac{64j}{2304k}} + 17$
 C) $l = \left(\frac{j}{k}\right)^2 + 17$
 D) $l = \left(\frac{64j}{2304k}\right)^2 + 17$

22. The given equation relates the distinct positive real number a, b , and c . Which equation correctly expresses a in terms of b and c ? (4)

$$\frac{35c}{5b} = 7\sqrt{a+12}$$

- A) $a = \sqrt{\frac{c}{b}} - 12$
 B) $a = \sqrt{\frac{245c}{35b}} - 12$
 C) $a = \left(\frac{c}{b}\right)^2 - 12$
 D) $a = \left(\frac{245c}{5b}\right)^2 - 12$

23. The formula below expresses s in terms of d, e , and t . Which of the following gives e in terms of the other variables? (4)

$$s = d^2 \sqrt{1 - \frac{e^2}{t^2}}$$

- A) $e = t \sqrt{1 - \frac{s^2}{d^4}}$
 B) $e = t \sqrt{1 + \frac{s^2}{d^4}}$
 C) $e = t \left(1 - \frac{s}{d^2}\right)$
 D) $e = t \left(1 + \frac{s}{d^2}\right)$

24. The equations below relate the distinct positive real numbers a, b, c , and d . Which equation correctly expresses b in terms of d ? (4)

$$c = 2a^2$$

$$a^{3b} = \left(\frac{c}{2}\right)^{d-1}$$

- A) $b = \frac{d}{3} + \frac{1}{3}$
 B) $b = \frac{4d}{3} - \frac{4}{3}$
 C) $b = \frac{2d}{3} - \frac{2}{3}$
 D) $b = \frac{2d}{3} - \frac{1}{3}$

25. The number of hours h for Saturn's moon Dione to complete an orbit around Saturn can be modeled by the equation below, where d is the average distance from Saturn, in thousands of miles. Which of the following expresses the distance in terms of the number of hours? (4)

$$h = 18 \cdot \sqrt[3]{\left(\frac{d}{67}\right)^4}$$

- A) $d = \frac{67}{18} (h)^{\frac{3}{4}}$
 B) $d = \frac{67}{18} (h)^{\frac{4}{3}}$
 C) $d = 67 \left(\frac{h}{18}\right)^{\frac{4}{3}}$
 D) $d = 67 \left(\frac{h}{18}\right)^{\frac{3}{4}}$